

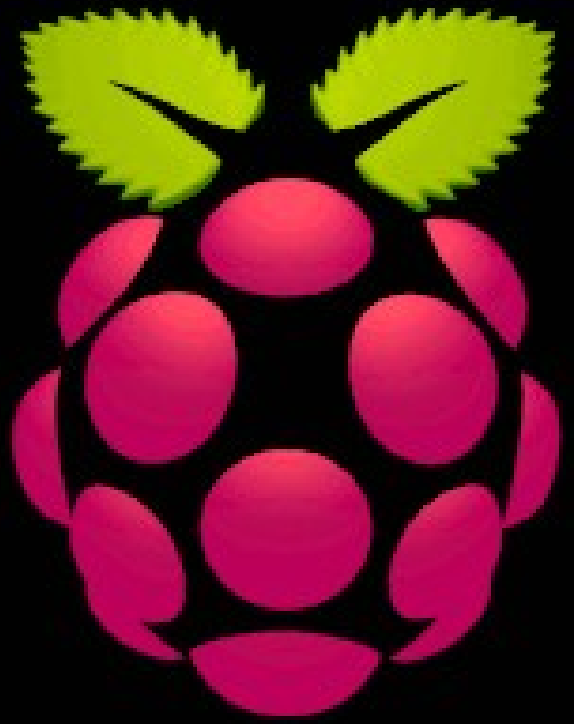
What is Under the Hood of a Modern Computer?

Traditionally we think of computers as the large desktop machines that many of us regularly use at home and at school.

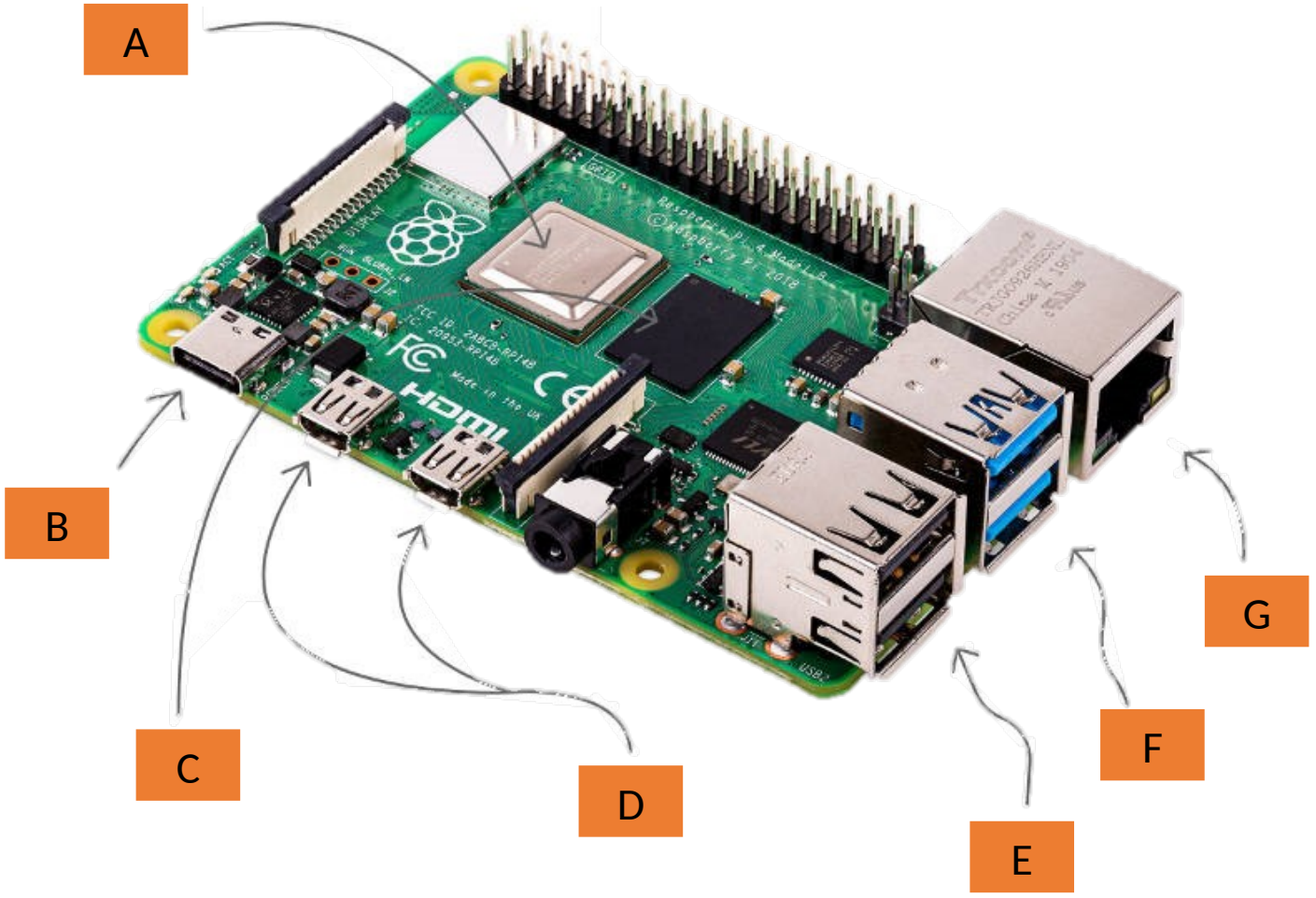
However, in recent years' computers have become faster and smaller

Raspberry Pi

One small computer is the raspberry pi, the credit-card sized computer that you can plug into the TV and a keyboard. Can you locate the processor, the memory, the storage and the input and output?



Label the Following diagram with the component parts:



Processor	?
USB-C Power Supply	?
USB 2	?
USB 3	?
Micro HDMI Ports	?
Ethernet Port	?
RAM	?

Key Terms



Input Devices: An input device enables the user to ‘input’ data into a computer.

Memory: This is where a computer keeps the data that has been input, as well as software applications and the results of any processing it has carried out, for the short term. This memory is lost when the computer is turned off.

Storage Device: This is where a computer stores files that have been created, as well as software that has been installed, for the longer term.

Processor: The part of a computer that processed data according to the instructions it has been given. It provides the user with information.

Output Device: An output device enables the user to receive information from a computer.

Homework: Spelling Test

Learn the following Key terms for a spelling test next lesson:

Folder
Backup
Graphics Card
Hardware
Employee
Repetitive Strain Injury
Password
Blind Carbon Copy
Input
Software
Raspberry Pi
Safety
Memory
Output
Motherboard
Eye Strain
Expectations
Hard Disk Drive
Processor
Attachment

UNIVERSAL

SERIAL

BUS

U S B

used to connect devices, transfer data, and supply power between computers and peripherals like keyboards, mice, printers



Understanding what makes your computer games console work

- https://www.bbc.co.uk/programmes/p07t308p



Algorithm



a *sequence* of instructions or a set of rules that are followed to complete a task or solve a problem. This task can be anything, so long as you can give clear, short and specific instructions for it.

An Example of an Algorithm

1: Come in and sit down quietly

2: Log on to your computer

3: Use a browser (internet explorer or google chrome) to navigate to google search engine

4: Research or find out what the term ‘algorithm’ means (You may need to visit a few sites)

5: When you have found out enough about the term ‘algorithm’ to explain its meaning to another student, turn off your monitor (screen) when you have finished.

What other examples are there?

- Recipe for baking a cake



INGREDIENTS

- 125ml coconut or rapeseed oil, plus extra for greasing
- 2 overripe bananas (about 200g flesh)
- 3 tbsp light coconut milk (plus 1 tbsp if using shredded coconut)
- 100ml clear honey
- 2 medium free-range eggs
- 1½ tsp vanilla extract
- 225g finely ground plain wholemeal flour
- 2 tsp baking powder
- ½ tsp chinese five-spice
- ¾ tsp ground cinnamon
- 1 tbsp shredded coconut (optional)

You will also need

- 900g loaf tin

METHOD

01. Heat the oven to 170°C/150°C fan/ gas 3½. Grease and line a 900g loaf tin. Using a fork or electric mixer, mash the bananas well.

02. Put the coconut milk, oil, honey, eggs and vanilla extract into a large bowl, then whisk together until combined. Stir in the mashed banana.

03. In a separate bowl, mix the flour, baking powder, five-spice, cinnamon and coconut (if using) with a fork. Sift these dry ingredients onto the banana mixture, then fold in using a metal spoon until just combined. Pour into the prepared loaf tin, then bake for 50-55 minutes until well browned and a skewer pushed into the centre comes out clean.

04. Leave to cool slightly in the tin for 5 minutes, then turn onto a wire rack. Leave for at least 15 minutes more or until cool before slicing.

Recipe taken from Asia Light by Ghillie James (RRP £15.99, Kyle Books)

What other examples are there?

- Directions to a specific place

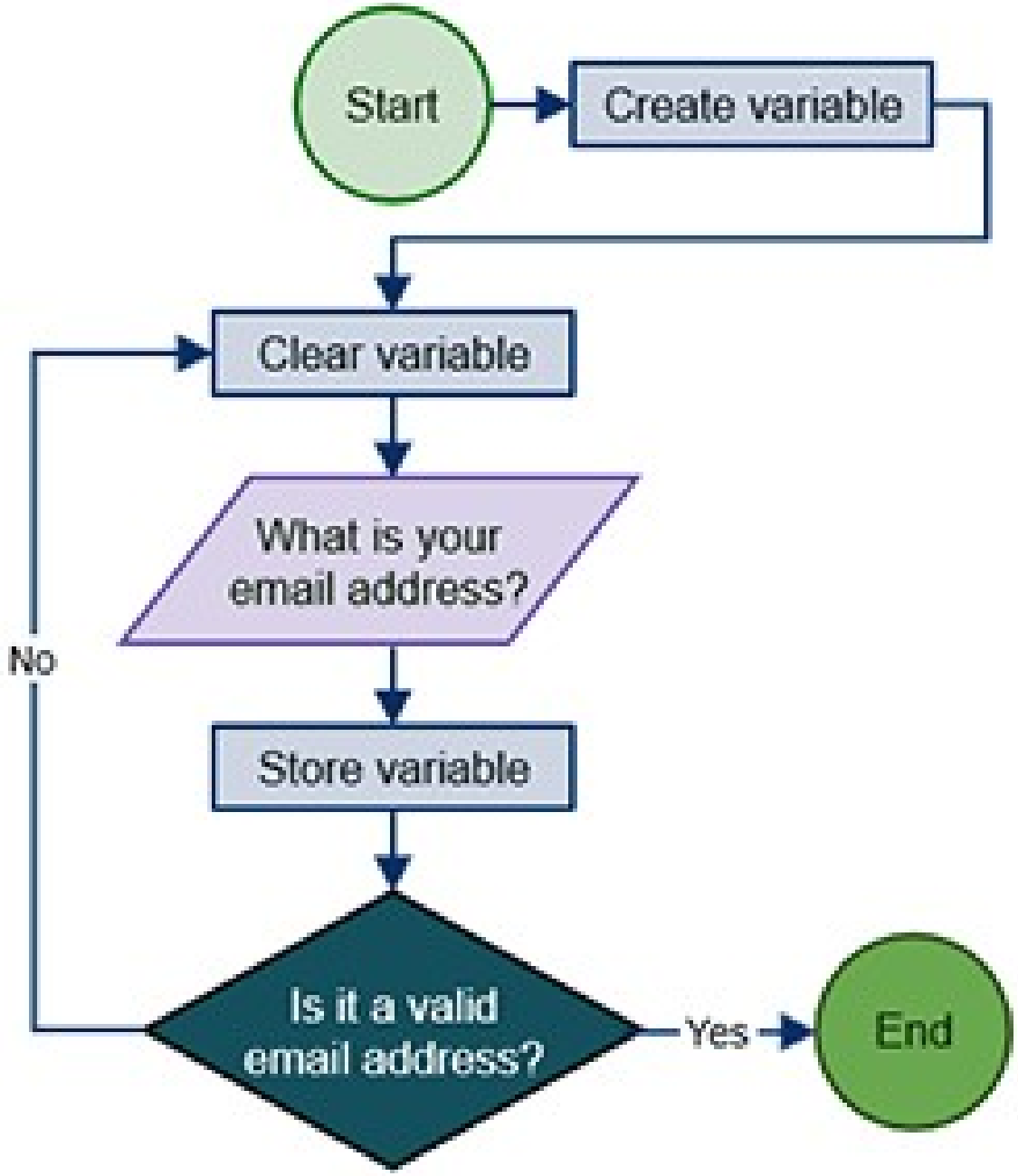


- *1: Turn right after the bank*
- *2: Walk straight until you get to the traffic lights*
- *3: Turn right*
- *4: Walk straight until you get to the traffic light*
- *5: Turn left*
- *6: Walk straight*
- *7: Follow the road round to the right (passing McDonalds on your left).*

- Computers are *very* good at carrying out sequence of precisely defined instructions.
- Algorithms can be used as a simply tool to help write a computer programs (which are simply instructions telling the computer what to do.)
- You may want a computer to calculate the answer to 25 divided by 5. Or tell you whether a certain number is prime or not.
- To more complex systems algorithms can be used to study the pages that you like on Facebook and determine which ads would be best to show you.
- Once you come up with a particular algorithm then you can implement it in any programming language of your choice.

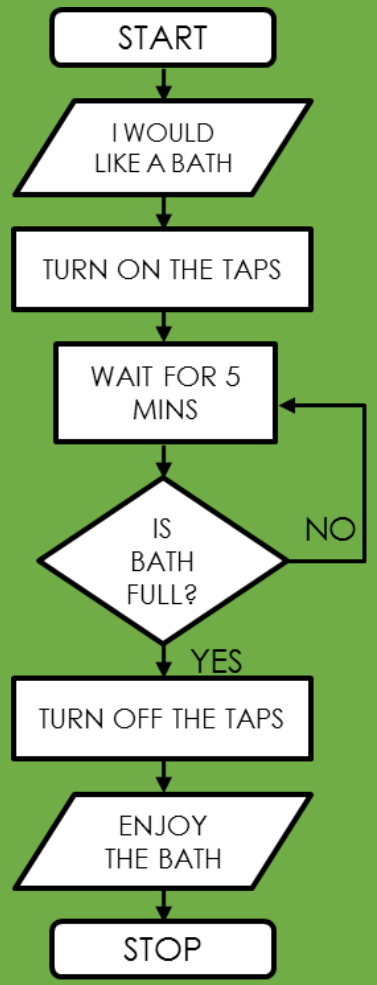
Computers & Algorithms

- Computer’s don’t actually understand algorithms as they use a different language.
- Algorithms need to be translated into code which the computer can understand and can follow to complete a task.
- Code is written in a programming language – just like speaking though there are many different programming languages.
- Some that you may come across are Scratch, Python and C#. Each of these languages are better suited to different things. Game developers use several different programming languages – sometimes even in the same game!



How computers carry out program instructions based on algorithms.

LEARNING TASK - Flowcharts
What is a flowchart?



A **flowchart** is made up of **symbols** and **arrows**.

A **flowchart** can be used to help design a **sequence**, which can control an event.

Actions can be **ordered**, **reordered** or **removed** if no longer necessary.

What is this flowchart about?

What are the different shapes?

What is unique about one of them?

LEARNING TASK - Flowchart Symbols
The four main symbols

START / STOP

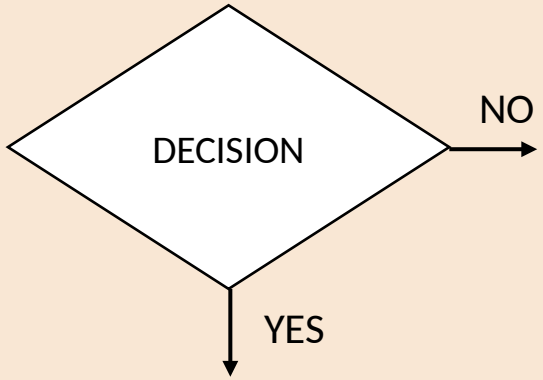
Start or stop a program

INPUT or OUTPUT

Receive data or give a signal

PROCESS/DELAY

Give an instruction; often involves a delay



Ask a question with a YES or NO answer

These are the main flowchart symbols you will need to know.

LEARNING TASK - Flowchart Symbols
The four main symbols

START / STOP

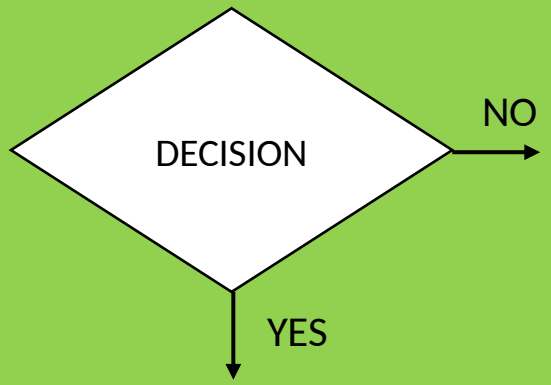
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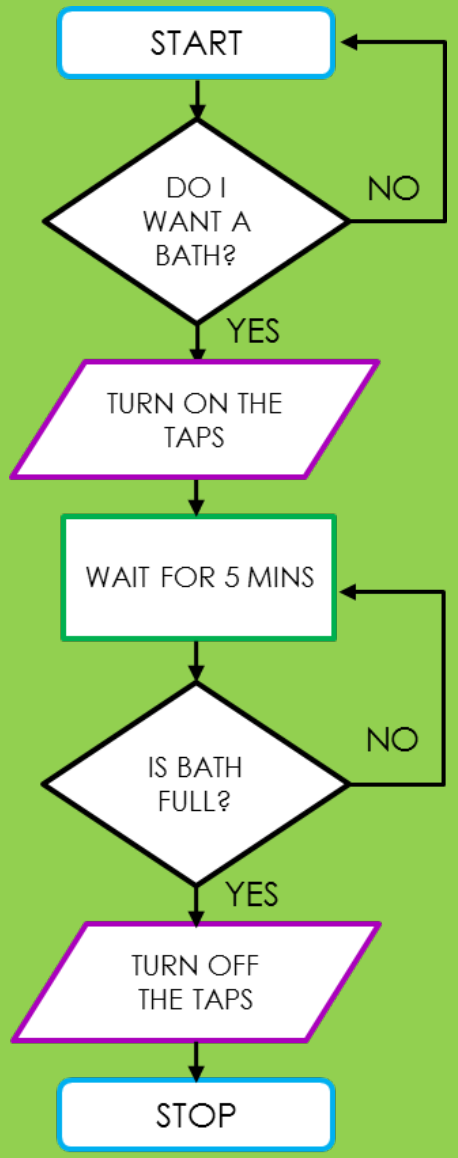
Receive data or give a signal

PROCESS/DELAY

Give an instruction; often involves a delay



Ask a question with a YES or NO answer



CLASS DISCUSSION - Which is better? Human vs. Computer

Which is **better**? Discuss…



vs.



Explain your reasons…

LEARNING TASK - Computer Control Advantages

COMPUTERS:

- ✓ do not need breaks.
- ✓ do not take sick days or holidays.
- ✓ can repeat actions in an identical way indefinitely.
- ✓ can work in environments that are dangerous for people.
- ✓ don't need to be paid.
- ✓ can process data from sensors very quickly.
- ✓ can manage hundreds/thousands of inputs at the same time.
- ✓ can make reliable and accurate decisions.

LEARNING TASK - Computer Control Disadvantages

COMPUTERS:

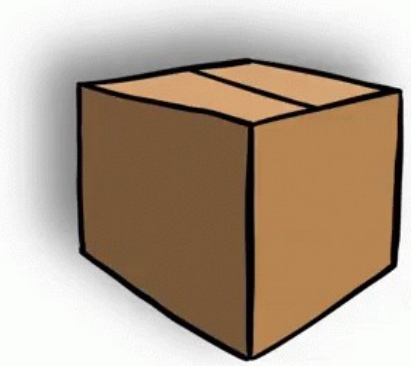
- X are only as good as their programming, which may contain errors.
- X that are faced with conditions not covered in their programming will not know what to do.
- X which are underpowered might take too long to do something in a time-critical situation.
- X rely on power; if there is a power cut the program will not work.
- X may cost a lot of money to purchase, maintain and develop.

How Computers Work

- Computers understand a very simple language – just 0 and 1s.
  - That's a lot less than the letters we use in our alphabet!
- This language is known as **binary**.
- Computers read millions of these 1s and 0s every second!
- Humans program these computers, often using **variables**. Follow the instructions on the next slide to see what this means!

What's a variable?

- A variable is a **location in memory** in which you can **temporarily store text or numbers**
- It is used like an empty box or the memory function on a calculator
- You can choose a name for the box (the "variable name") and change its contents in your program.



X = 50

X = X + 5

X is the variable



Hair Colour (Black/Blue/Blonde)

Piercings (One/Both/Style)

Hair style (Long/Short/Combed/Frizzy)

Glasses (Yes/No).

Tattoos (What/Where/Colour)

Choice of Shorts (Long/Colours)

Choice of Footwear (Trainers/Boots/Shoes)

Choice of Top (Tank/T Shirt/Shirt)

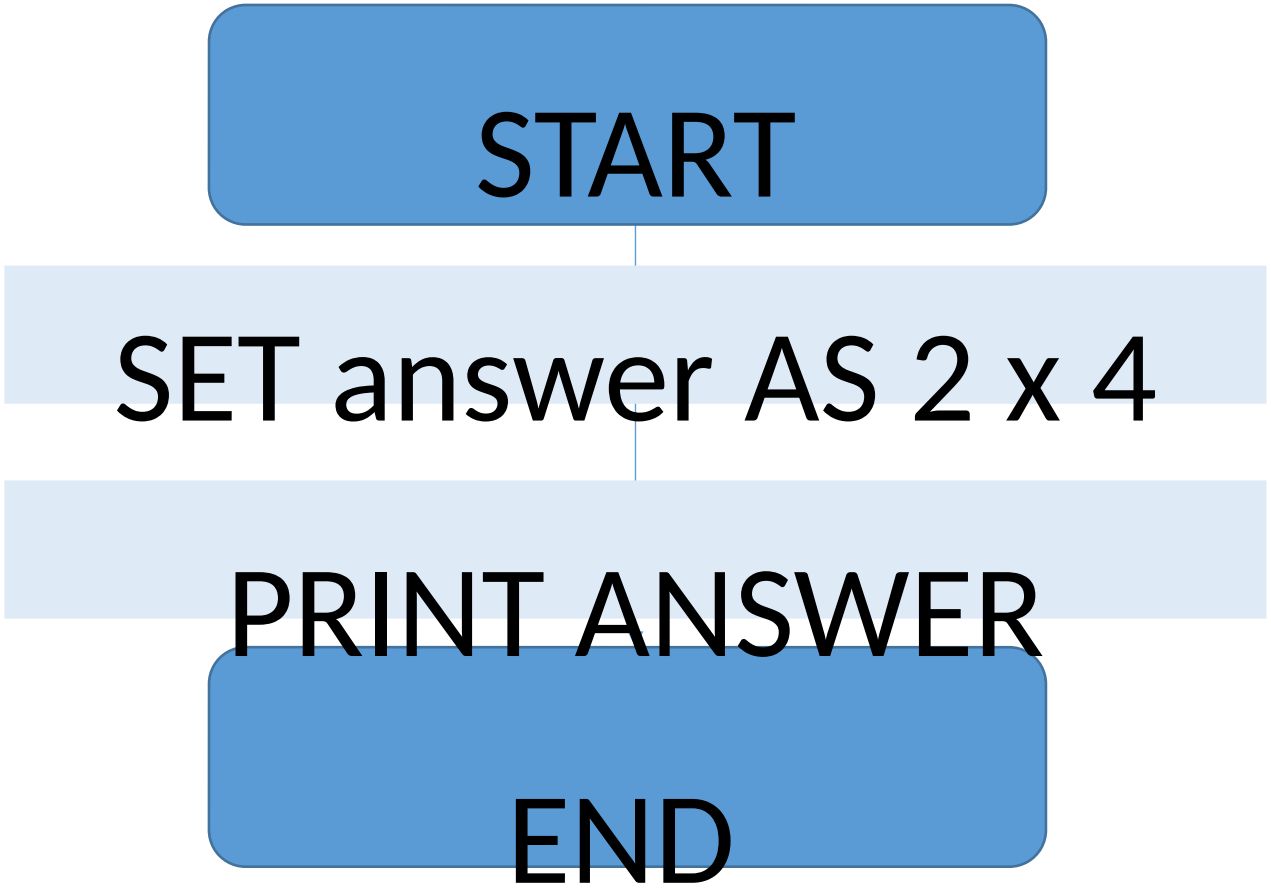
Name? Age? Interests? There are literally thousands of variables!

So here is a ‘human’ object. The question is what VARIABLES are there for this human? What makes them different from one another?

Variables

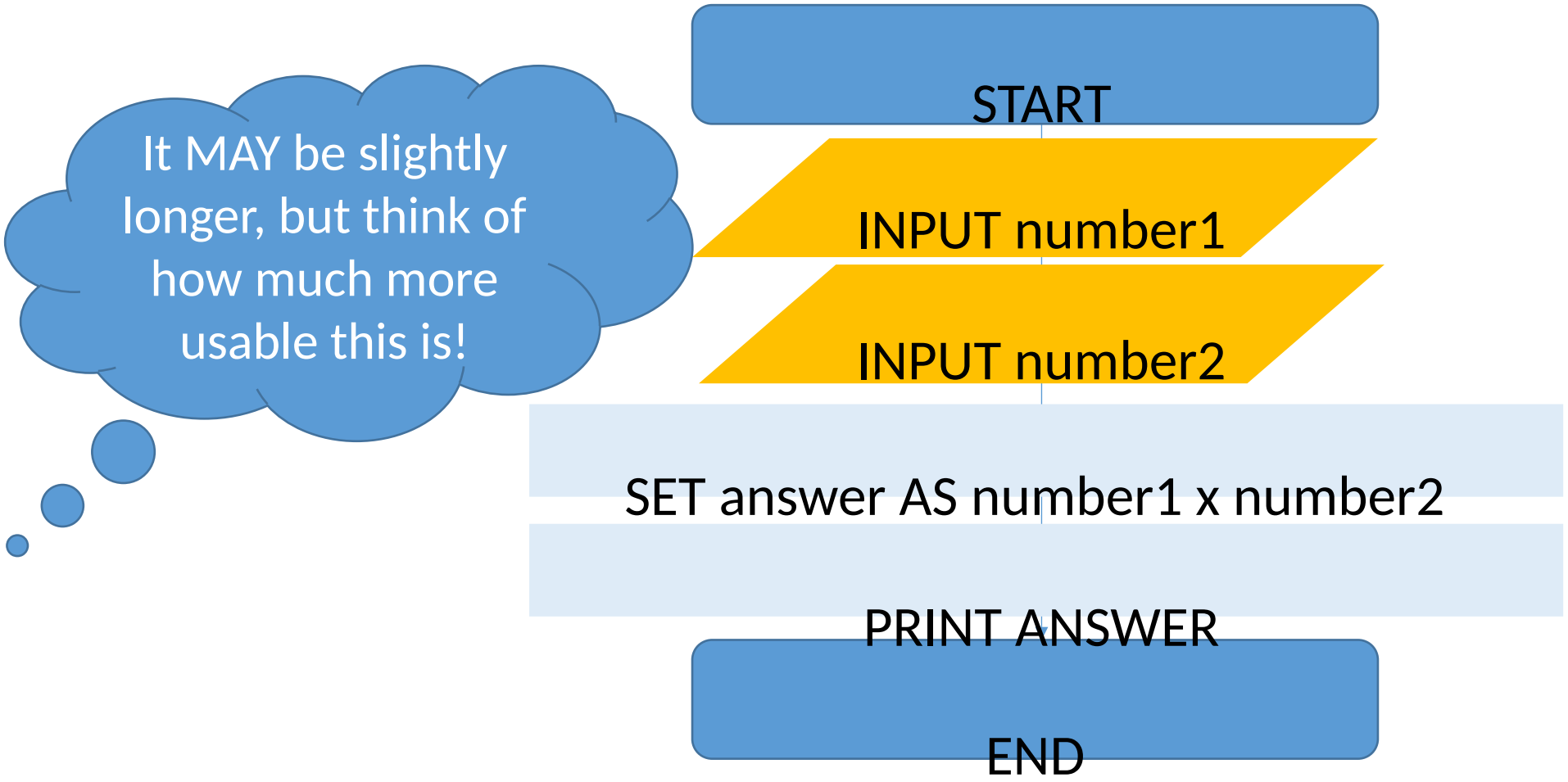
A simple Program

- Well, We could have a program that works as a calculator!



So, how does this help?

- Well, We could add **variables** in order to make this more usable



Variables

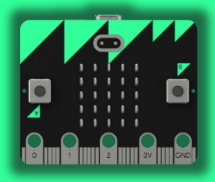
- Variables make our programs a lot more reusable. Simple things like displaying your name at the top of the screen, or having a score appear.

- We can store text, numbers and even ‘true’ or ‘false’ values

- Variables allow us to **store, change** and **access** these values as the program runs.

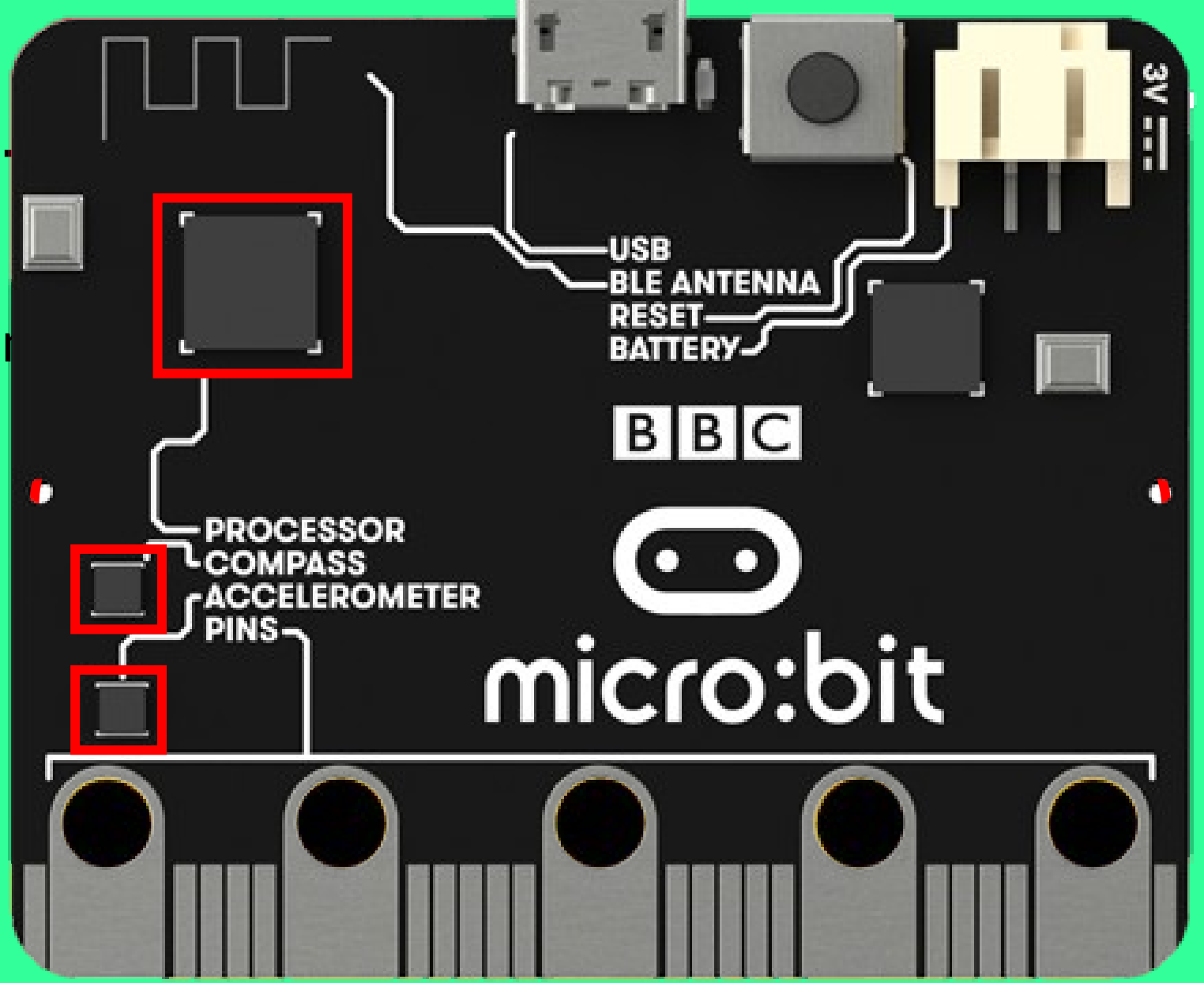
What is the BBC Micro:Bit?

- Pocket Sized codable computer
- 25 red LEDs that can light up, flash messages, create games and invent digital stories
- Two programable buttons.
  - Can use it as a games controller
  - Can even be used to pause or skip songs on a playlist!
- Has a motion detector (accelerometer) that can detect movement
- Built in compass (magnetometer)
- Bluetooth smart technology – can connect to other Microbits or devices



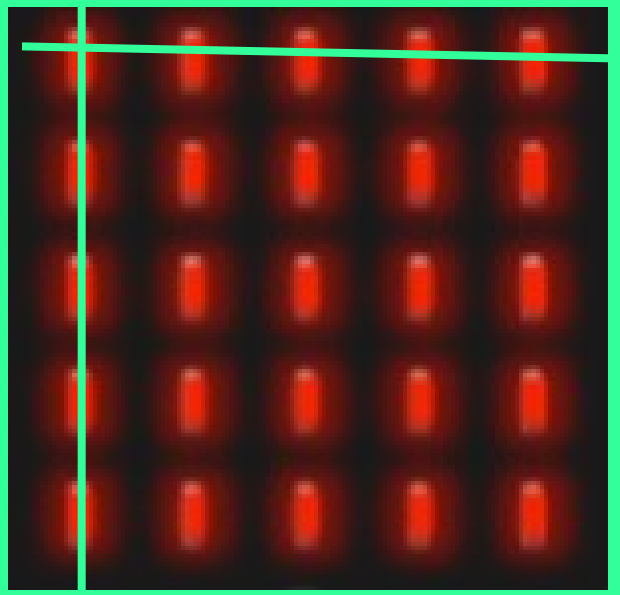
The Componen

- Programmable Buttor
- 25 LEDs
- Compass
- Bluetooth
- Accelerometer



How does the LED light array work?

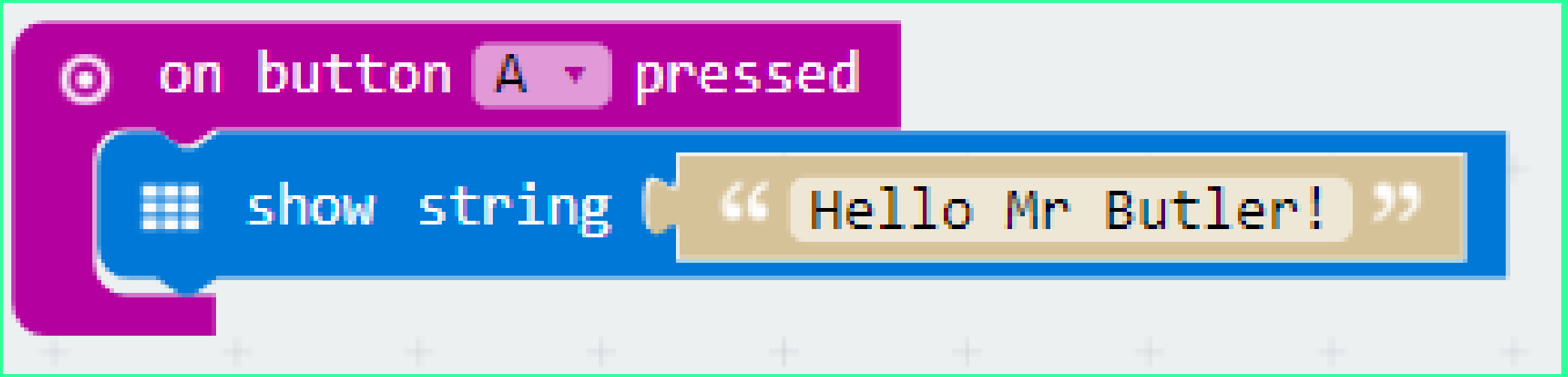
- The LEDs are arranged in a 5x5 grid
- Much like coordinates, these can be referenced to X and Y.
- Unlike coordinates, we reference them like so:



Programming Buttons

- With 2 programable buttons, we can use these as inputs for our Microcontroller.

- Let’s make a short program:



Looping

- Q: What does the word ‘Loop’ mean in programming?
- Ans: To repeat.

- For example, we could loop 2 or 3 times. Or 5. Or 100. Or even **forever**



